# Micro-Expression Motion Magnification: Global Lagrangian vs. Local Eulerian Approaches

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***Abstract*—Micro-expressions are difficult to spot but are utterly important for engaging in a conversation or negotiation. Through motion magnification, these expressions become much more distinguishable and easily recognized. This work proposes Global Lagrangian Motion Magnification (GLMM) for consistent exaggeration of facial expressions and dynamics across a whole video. As the proposal takes an opposite approach to a previous pivotal work, i.e. local Amplitude-based Eulerian Motion Magnification (AEMM), GLMM and AEMM are theoretically analyzed for potential advantages and disadvantages, especially with respect to how magnified noise and distortions are dealt with. Then, both GLMM and AEMM are empirically evaluated and compared using the CASME II micro-expression corpus.**

***Keywords*-Micro-expressions, motion magnification, Lagrangian, Eulerian, multi-channel gradient model**

## I. Introduction

There are hundreds [3] of facial muscles which we learn to control precisely, in order to express or suppress our emotions. This makes facial expression a vital visual medium for conveying our emotional states to other human beings. We can almost effortlessly pick up these emotional clues, as our abilities in recognising normal expressions are sharpened through daily encounters. However, there is a type of facial expressions that is proving to be extremely challenging for an untrained person [4]. These are referred to as micro-expressions. These expressions are extremely hard to recognise as they are involuntarily and generated suddenly with very small intensity in a fraction of time (i.e. between $\frac{1}{25}$s and $\frac{1}{3}$s). A few video samples of spontaneous micro-expressions are displayed in Figure 1 as a micro-expression recognition challenge for readers. In a psychological experiment, Frank et al. [4] report a 32% success rate in correctly recognising five types of micro-expressions for an untrained person. When one is trained with Micro-Expression Training Tools (METT) by Paul Ekman's Group, one has a 45% chance of being correct in this 5-class recognition task. As well as being challenging for human beings, this task has also proven to be difficult for machines. Le-Ngo et al. [9] report 44% accuracy for a basic machine learning system. Importantly, a Leave-One-Subject-Out (LOSO) protocol is utilised in evaluation steps of [9]. This standard protocol [18] provides more consistent evaluation results than the Leave-One-Video-Out (LOVO) subject, used in a white paper of the CASME II database [20]. The challenge of automatic recognition of micro-expressions has recently attracted considerable attention and research effort from both the psychology and computer-vision community. Several standard corpora e.g. CASME II [20], CASME [21], and SMIC [11] have been acquired through carefully designed experiments. Importantly, the recorded videos containing micro-expressions, are post-processed by METT-trained experts for reliable annotation and ground-truth. Equipped with these high quality databases, several works, i.e. Le-Ngo et al. [10], Oh et al. [15], Liong et al. [12], Huang et at. [5], etc. investigated various approaches for improving the performance of machine learning systems in an attempt to solve this challenge. A majority of these works were aimed towards introducing more comprehensive and effective features, e.g. Spatial Local Binary Pattern with Integral Projection [5], Monogenic Riesz-Wavelet Representation [15], etc.

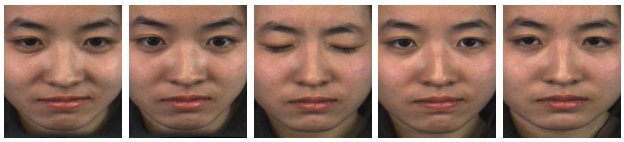



Figure 1. Five types of micro-expressions from the CASME II corpus. Only representative frame of these video sequences are provided.

A few recent works [7][8][10] focused on manipulating dynamics of facial expressions to make them more discriminant for machine learning systems. For example, amplification of the intensities in facial expressions has been previously investigated by Park et al. [16], [17], Le-Ngo et al. [8]. Park et al. [16] estimated the optical flow at facial feature points, detected using the Active Appearance Model (AAM). Then, they magnified the motion vectors between corresponding fiducial points of two consecutive frames before warping normal expressions into exaggerated ones. Furthermore, Park et al. [17] and Le Ngo et al. [8] adopted Amplitude-based Eulerian Motion Magnification (AEMM),

developed by Wu et al. [19] to magnify the emotion although each work has a different strategy for the employment of magnification factors. While Le-Ngo [8] surveys across a wide range of magnification factors, Park [17] adapts magnification w.r.t. the temporal dynamics of specific micro-expressions. Another strategy for modifying facial dynamics was the compression of the dynamics of micro-expressions in time, which was successfully employed by Le-Ngo et al. [10] to improve recognition rates.

This research paper falls into the latter theme as it proposes Global Lagrangian Motion Magnification (GLMM) to visually exaggerate micro-expressions and subsequently increase their discrimination. Despite having a similar spirit, GLMM and AEMM are different in the following aspects of how motion is computed (i.e. Lagrangian vs. Eulerian frameworks), processed (i.e. globally in GLMM and locally in AEMM) and how magnified noise and distortions are handled. AEMM is briefly summarized in Section II, then Section III elaborates and differentiates the proposed GLMM against AEMM. The proposed method is evaluated in a framework of subtle emotion recognition on the CASME II corpus, described in Section IV. Meanwhile, Section V lists and compares performances of GLMM and AEMM in tables and plots. Finally, Section VI summarises advantages and disadvantages of the considered methods as well as our future research direction.

## II. Eulerian Motion Magnification

Amplitude-based Eulerian Motion Magnification (AEMM) exaggerates differences in the brightness or colour of pixels at the same spatial location across two consecutive frames. Mathematically, AEMM is written as follows.

$$I(x, y, t+1) = I(x, y, t) + \sum_k \sigma_k \frac{\partial I(x, y, t)}{\partial t} \tag{1}$$

where $I, x, y, t$ represent an intensity, spatial locations and a temporal index of a pixel respectively. In AEMM, an image is first decomposed into multiple levels $k$ of dyadic scale space before each of the decomposed signals is magnified with an appropriate magnification factor $\sigma_k$. At each level $k$, $\sigma_k$ is computed from the input magnification factor $\sigma$ so that it reduces distortion from the magnification process. The derivation of $\sigma_k$ depends on a very simplistic assumption about signals (i.e. variation of pixel intensity is modelled by a smooth cosine function) and heuristic scale-space parameters. Therefore, AEMM risks distorting edge-like or more complicated image features if the scale-space smoothing windows cannot cover changes of these features through time. More details about AEMM and its magnification factors can be found in [19].

## III. Lagrangian Motion Magnification

Unlike the Eulerian view, the Lagrangian approach requires explicit tracking of all displacements through space and time. In the Lagrangian framework, each pixel can be traced back and forth along its pathline, a locus of its positions, through time. Given a displacement in the 2-D plane $(u, v)$, every pixel can be temporally traced to its previous position. And thus too a frame, a collection of pixels as follows.

$$I_{t+1}(x, y, t+1) = I_t(x+u, y+v, t) \tag{2}$$

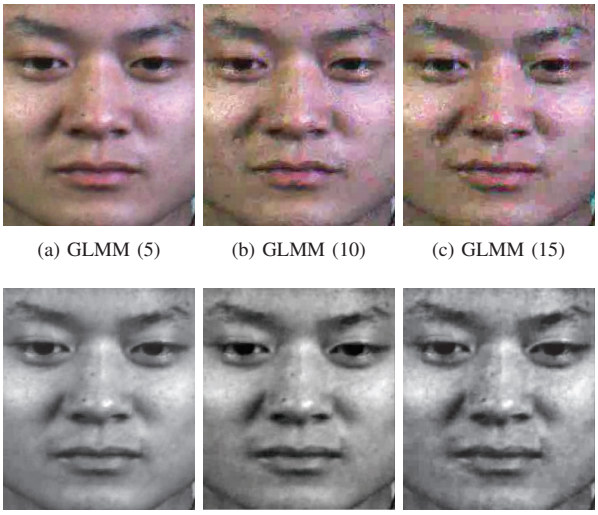
Displacements can be observed locally or globally. Lets consider the movements of a point through spatial locations A, B and C at time $t, t+1, t+2$. Locally, its movements are described by vectors $\overrightarrow{AB}, \overrightarrow{BC}$ as they are observed w.r.t. some previous location. Alternatively, these movements can be written as $\overrightarrow{OB} - \overrightarrow{OA}$ and $\overrightarrow{OC} - \overrightarrow{OB}$ in which $O$ is a common reference point. Utilisation of a reference point is a convenient way to describe all movements regardless of previous locations. It gives a global Lagrangian view of motion. A similar view can be applied to pixels in video frames in which global displacement is estimated between target $I_T$ and reference $I_R$ frames. Given the displacement is specified by $(\mathbf{u}, \mathbf{v})$ in 2-D, a target frame can be recovered from a reference frame by a warping operator. Furthermore, a target frame can be magnified by multiplication of some displacement with magnification factors $\sigma$

$$I_T = \text{warp}(I_R, \sigma(\mathbf{u}, \mathbf{v})) \tag{3}$$

In this proposed Global Lagrangian Motion Magnification (GLMM), $(\mathbf{u}, \mathbf{v})$ is estimated by the Multi-channel Gradient Model (McGM) [6]. In addition, a warping technique is adopted from the research of Johnston and his associates [2], [14] rather than simply mapping pixel value as it produces a sharper caricature of facial expressions. Moreover, the technique synthesizes more realistic facial dynamics of different micro-expressions; further details about McGM and the warping operator can be found in [2], [13]. The global Lagrangian approach allows learning significant motion from all the dynamics and noise in a video. In GLMM, Principal Component Analysis (PCA) is employed to learn statistically dominant displacements for all video frames. Furthermore, truncation of small principal components potentially filters out insignificant movements, and noise while retaining meaningful movements. In the experiments described below, GLMM uses less than 20 of the most significant modes, ignoring the rest. The filtered displacement is approximated as below.

$$(\mathbf{u}, \mathbf{v}) \approx \sum_1^k \mathrm{C}_k \mathrm{PC}_k(\mathbf{U}, \mathbf{V}) \tag{4}$$

where $k$ in GLMM means the $k$-th principal component $\mathrm{PC}_k$ of the global collection $(\mathbf{U}, \mathbf{V})$ with correspondent coefficients $\mathrm{C}_k$ of displacement vectors $(\mathbf{u}(\mathbf{t}), \mathbf{v}(\mathbf{t}))$. After



(a) GLMM (5) (b) GLMM (10) (c) GLMM (15)

(d) AEMM (5) (e) AEMM (10) (f) AEMM (15)

Figure 2. Motion magnification of a sample CASME II sequence by GLMM (on first 9 principal components) (a, b, c) and AEMM (d, e, f) with their magnification factors shown in brackets. Only the representative frame of these magnified sequences are provided. The magnified video sequences are available for viewing at the project website[1].

substituting $(\mathbf{u}, \mathbf{v})$ of Eqn (4) into Eqn (3), we can summarise the proposed GLMM as follows:

$$I_T = \text{warp}(I_R, \sigma \sum_{1}^{k} \text{C}_k \text{PC}_k(\mathbf{U}, \mathbf{V})) \tag{5}$$

For a closer look, we examine a few magnified expressions by applying AEMM and GLMM with various magnification factors i.e. 1, 5, 10, 15, 20 on the video sample 'EP02_01f' of male subject '01' from the CASME II corpus. When magnification factors are under 10, both methods give similar exaggeration of facial expressions. Differences only become noticeable with $\sigma = [5, 20]$. Note also that, GLMM and AEMM suffer from different types of distortions at higher magnification factors i.e. 15–20. For example, unnatural change in brightness are seen in the cheek areas of the subject in AEMM, while GLMM is unable to recover some pixels due to amplification of noise in the same area.

Subjectively, the videos (provided as supplementary material) of the samples in Figure 2 provide the evidence that GLMM gives more discriminant magnification than AEMM does across a variety of magnification factors. Objectively, quantitative evaluations are carried out within a subtle emotion recognition framework to numerically compare their performances, as described in the next section.

## IV. Subtle Emotion Recognition

CASME II is a comprehensive database of recorded video samples of spontaneous subtle emotions. Specifically, it contains 247 samples, elicited from 26 Asian participants with an average age of 22.03 years old. During the database acquisition, human participants were asked to suppress their expression while they were deliberately stimulated by a series of short videos. These videos were selected to evoke five categories of micro-expressions: Tense (T), Disgust (D), Happiness (H), Surprise (S) or Repression (R). Recordings of participants' expressions were then post-processed by METT-trained specialists to identify the location and length of the occurrences, and the category of the micro-expressions. The specialists had access to high frame-rate videos i.e. 200 $fps$ at $280 \times 340$ resolution from a Point Grey GRAS-03K2C camera since micro-expressions are naturally elusive and short-lived. To improve the fidelity of the high-speed recording, four LED lamps under umbrella reflectors were used to fill the recording room with stable lightning conditions to avoid flicker. In short, CASME II is a carefully designed corpus of spontaneous subtle expressions with high-quality video samples, reliable ground-truth and a reasonable numbers of subjects.

Amplitude-based Eulerian Motion Magnification (AEMM) and Global Lagrangian Motion Magnification (GLMM) were applied to videos of cropped inner faces. As each video sample in CASME II has slightly varying resolution, they were all re-sized to a fixed resolution 340x280 as suggested in the white paper of the CASME II databases [20]. Then, each video was converted from color to gray-scale from which LBPTOP features [22] were extracted with the following parameters: [4,4,4,1,1,4]. The first three parameters denote that center-surrounding comparisons are with four neighboring pixels in four directions (i.e. North, South, East, West) for all three orthogonal planes (XY,XT,YT), while the last three parameters specify distances between central and surrounding pixels (e.g. 1 and 4 denote that comparison is made between the central pixel with the 1st and 4th nearest pixel respectively). Note that LBPTOP features were not extracted from the whole frames but rather from spatio-temporal blocks of 5x5 checkerboard partitioned frames. Specially, a universal rotation-invariant version of LBPTOP was utilized; hence, each block generates a 49-D feature vector. These vectors were later concatenated into a 1125-D vector (i.e. 25 blocks x 49-D ), a representative feature for each video in the later machine learning (ML) process. AEMM and GLMM were evaluated on the CASME II corpus for a range of magnification factors $\sigma \in [1 : 20]$. In addition, GLMM magnifies the corpus with the number of principal components $1 \leq PC \leq 20$, used for reconstruction of exaggerated expressions. Therefore, supervised training and testing processes were carried out for 20 configurations of AEMM and $20 \times 20$ configurations of GLMM. In each configuration, a Support Vector Machine (SVM) with a Linear Kernel (c = 10000) is used for evaluating how effectively the proposed motion magnification techniques

[1] http://pesona.mmu.edu.my/~johnsee/research/microexp

Table I
Summary of parameters & configurations

| Corpus | | | |
|---|---|---|---|
| CASME II | | | |
| Subject | No of Samples | Classes | FPS |
| 26 | 259 | 5 | 200 |
| Motion Magnification | | | |
| AEMM | | GLMM | |
| x | PC | x | PC |
| [1-20] | N/A | [1-20] | [1-20] |
| Feature Extraction | | | |
| Image Size | Colour | Universal LBPTOP | |
| 340x280 | Gray | [4,4,4,1,1,4] | 5x5 blocks |
| Machine Learning | | | |
| Type | Kernel | Configuration | Protocol |
| SVM | Linear | C=10000 | LOSO |
| Evaluation Measurement | | | |
| F1 | Recall | Precision | Accuracy |

improve recognizability of facial micro-expressions. In this study, the constrain parameter (c) is set extremely large to reduce mis-classification in the training stage arising from identities between samples of the same subject. In addition, a leave-one-subject-out (LOSO) protocol is utilized for elimination of errors with respect to overlapping subject identities between training and testing sets. In brief, LOSO consecutively selects videos belonging to a subject out the corpus as a test set while the rest are used in the corresponding training set. The final result is an average evaluation with each subject employed once as a test subject. As CASME II has 26 subjects, the LOSO protocol dictates SVM runs and evaluations on 26 different training and testing sets. In regard to specific statistical measurement in each evaluation, F1 scores, recall and precision rates are employed instead of overall accuracy since these measures provide a fair evaluation in an imbalanced corpus like CASME II [9]. All important details of experiments in this paper are summarized in Table I.

## V. Experiments and Analysis

The proposed GLMM method is evaluated in the micro-expression recognition framework with 400 different configurations corresponding to the various magnification factors ($\sigma \in [1:20]$) and principal components PC ($\in [1:20]$). The evaluation result i.e. F1 score is then organized into a 20x20 table. As the measurements vary w.r.t. both variables, two box-plots in Figs. 3 & 4 help to visualize and analyze performances w.r.t. each variable separately. Each box in the box-plots graphically depicts statistics of GLMM's performances w.r.t. a range of configurations (e.g. a range of PC in Fig. 3 and a range of $\sigma$ for Fig. 4). Upper and lower boundaries of boxes together with two whiskers at both extremes equally divide sorted data into four quartiles, each of which contains 25% of the total number of measures. To compare with GLMM, we evaluate AEMM over the similar magnification factor range i.e. [1:20] and the results are plotted in red in the same figures (Figs. 3 and 4). Furthermore, Table II concisely

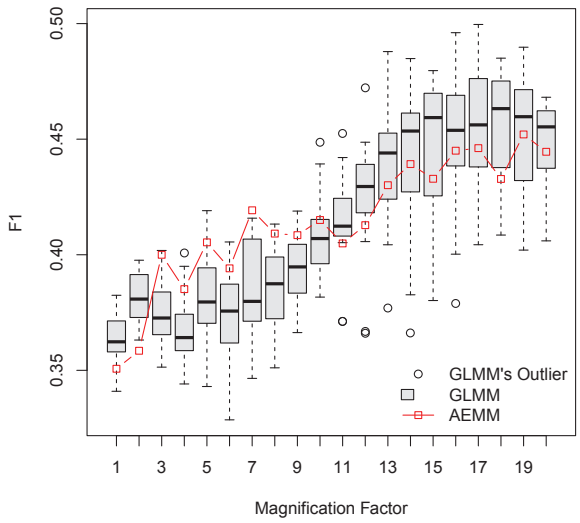



Figure 3. Performances of GLMM and AEMM w.r.t. Magnification Factor $\sigma \in [1:20]$. For each $\sigma$, GLMM generates 20 F1-scores w.r.t. PC $\in [1:20]$, which are box-plotted to depict their quartiles. Outlier results for corresponding box-plots are denoted with $\circ$ symbol.

summarizes the comparison between AEMM and GLMM by listing both of maximum and minimum performances along with their specific experimental configurations.

GLMM claims to be more efficient at filtering out noise and only keeping statistically dominant facial movements (see Section III). As significant principal components of global displacements are learnt by PCA and the residue is truncated, less noise is present in the later magnification processes. The comparison of performances between GLMM and AEMM at magnification factor 1 (i.e. no magnification factor) in Figure 3 supports the hypothesis. The figure shows the F1-scores for GLMM vary from 0.34093 to 0.38246 w.r.t. the different principal components. Its median is 0.36, which is higher than corresponding F1-score of AEMM i.e. 0.35. Though the difference at $\sigma = 1$ seems to be insignificant (0.01 of F1-score), its effect becomes increasingly obvious for magnification factors larger than 10. Interestingly, GLMM's median performance is worse than that of AEMM only at PC = [16,17,18]. This shows that not all dynamics of facial expression help to discriminate these types of subtle emotion. It is therefore advantageous to use only the first 15 dominant principal components of the whole facial dynamics in each CASME II video sample without magnification.

With increasing magnification, the performances of both GLMM and AEMM increase in tandem. Figure 4 shows that when more than 1 or 2 PCs are utilized in GLMM, its performance is better or at least equivalent to AEMM

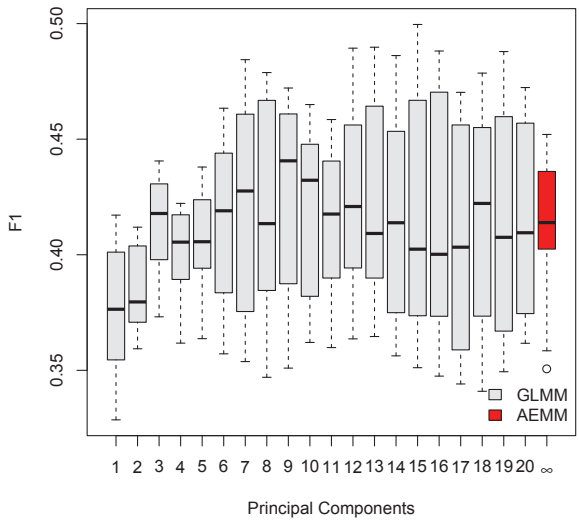



Figure 4. Performances of GLMM and AEMM w.r.t. PC $\in [1:20]$. For each principal component index (PC), GLMM uses all principal components from 1 to PC to reconstruct exaggerated motion (denoted by summation over $k$ variable in Eqn. 4).

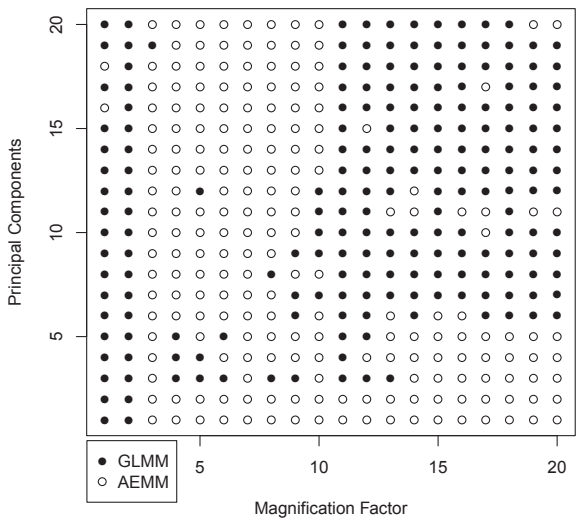



Figure 5. Go-Chart of GLMM and AEMM over ranges of parameters (PC, $\sigma$). If GLMM performs better than AEMM, it wins that position. Otherwise, AEMM wins.

Table II
Overall performance of GLMM vs AEMM

| Measure | GLMM (PC, $\sigma$) | | AEMM ($\sigma$) | |
|---|---|---|---|---|
| | Min ( 1 , 6 ) | Max ( 15 , 17 ) | Min ( 1 ) | Max ( 19 ) |
| F1 | 0.33 | 0.50 | 0.35 | 0.45 |
| Recall | 0.32 | 0.47 | 0.33 | 0.41 |
| Precision | 0.34 | 0.54 | 0.38 | 0.50 |

across various magnification factors. Interestingly, there is at least one principal component (PC) at each magnification factor where GLMM performs better than AEMM. Medians of GLMM's F1-scores across all PC values are above those of AEMM for 12 magnification factors i.e. [1-2, 11-20]. At the range $\sigma$=[1,2] (very small magnification factors), GLMM is better than AEMM as it picks up significant dynamics and eliminates noise. With magnification factors between 2 and 10, AEMM outperforms the medians of GLMM as shown in Figure 3 but not the best F1-score of GLMM at each magnification factor. However, GLMM is better than AEMM at both maximum and median performances for $\sigma$=[11-20] (larger magnification factors). Also, GLMM supports a wider range of magnification factors than AEMM. As the proposed method only picks up significant dynamics, it only magnifies these dynamics. Meanwhile, AEMM magnifies all changes regardless of whether this is due to significant movement or noise as Wu et al. [19] linearly vary magnification factors $\sigma_k$ at each pyramid level rather than totally truncating negligible components. Therefore, AEMM is much more affected by magnified noise than GLMM at large magnification factors. Figure 5 shows where in configuration spaces that GLMM is better than AEMM. Statistically, GLMM and AEMM are equivalent in the number of configurations where one is better than the other. However, their dominant configurations are distributed in a different manner. GLMM is dominant at top-right corner of the Go-Chart in 5 i.e. higher magnification factors and principal components in their respective ranges. Interestingly, the general trend seems to be reversed for the smallest two principal components and magnification factors. For instance, GLMM performs better than AEMM in extremely small factors (i.e. 1, 2) across almost all components, whereas AEMM dominates in high magnification factors when only the first two principal components are considered. There are several works in recent literature that are aimed towards better feature extraction for micro-expression recognition [5], [12]. Instead of LBPTOP, Huang et al. [5] utilized horizontal and vertical integral projections of the difference between two consecutive frames in their proposed feature extraction technique named Discriminative Spatio-temporal Local Binary Pattern Improved Integrated Projection (DiSTLBP-IIP). Meanwhile, Liong et al. [12] proposed the Bi-Weighted Oriented Optical Flow (Bi-WOOF), which estimates the optical flow between the apex frame and first (neutral) frame to summarise the facial activities in a video sample. Ultimately, both these methods then encode their respective extracted information using the Local

Table III
Performance comparison against state-of-the-art methods on CASME II

| Measure | LBPTOP | | DiSTLBP-IIP | Bi-WOOF |
|---|---|---|---|---|
| | Baseline [20] | GLMM | [5] | [12] |
| F1 | 0.39 | 0.50 | 0.62 | 0.61 |
| Recall | 0.38 | 0.47 | 0.60 | 0.60 |
| Precision | 0.41 | 0.54 | 0.65 | 0.63 |

Binary Pattern descriptor [1] to form feature vectors for classification.

As shown in Table III, both of the mentioned methods significantly outperform the "Baseline" LBPTOP feature first reported by Yan et al. [20] with CASME II. This big difference in performance is due to the suitability of these feature extraction techniques for CASME II video samples, which contain faces that are well illuminated and carefully cropped and aligned in advance. LBPTOP as a universal feature descriptor may not be designed to cope with environmental changes such as flickering brightness, slight temporal transitions, etc. To make things worse, LBPTOP is likely to suppress tiny but useful facial movements as it is sensitive to the local neighborhood that it operates in. Meanwhile, the other compared methods (DiSTLBP-IIP and Bi-WOOF) were able to successfully extract and encode the important dynamics since they perform some sort of reduction in spatial (integral projection) and temporal information (optical flow) that requires faces to be well-aligned. As such, the DiSTLBP-IIP and Bi-WOOF features might not necessarily perform better in the case of less well-conditioned faces. Hence, more thorough evaluations on micro-expression videos collected under less controlled environments are necessary to confirm their advantages. In this GLMM approach, motion magnification is proposed as a pre-processing step, totally independent from feature choices. The first two columns of Table III i.e. LBPTOP-Baseline and LBPTOP-GLMM show that significant improvement can be obtained by separately employing GLMM before LBPTOP. Similar increase in performance is expected if GLMM is utilized before other features are extracted with methods such as DiSTLBP-IIP and Bi-WOOF. While the encoding of facial dynamic features through spatial integral projections or temporal estimation of optical flow information have proven to be quite an effective solution, future integration of motion magnification with these robust feature extractors would provide further boost to the overall recognition performance.

## VI. Conclusion and Future Work

In summary, this work introduces the Global Lagrangian Motion Magnification (GLMM) for micro-expression magnification. It is an entirely opposite approach to the previous techniques as GLMM considers dynamic change in terms of the global displacement between a target and a reference frame instead of the local displacement between consecutive frames. Furthermore, it applies PCA to filter out trivial components of the dynamics, therefore less noise and insignificant movements are magnified. Overall, GLMM produces consistent magnification of expressions across larger ranges of magnification factors than AEMM does. It also produces better recognition rate (F1-score) than AEMM for almost all magnification factors. Though GLMM performs better than AEMM in CASME II, the evaluation only focuses on average performance across different classes of micro-expressions. Further work would be analysing in details on how GLMM performs for each micro-expression class.

## Acknowledgment

The authors would like to thank the anonymous reviewers for their helpful and constructive comments towards the improvement of this paper. Prof Johnston is funded by the EPSRC (EP/M026965/1) and the NIHR Nottingham Biomedical Research Centre. RP gratefully acknowledges the support by the EPSRC under the project Signal Processing Solutions for the Networked Battlespace (EP/K014307/1∼2) and the MoD University Defence Research Collaboration (UDRC) in Signal Processing. JS's work was supported in part by the MOHE Grant FRGS/1/2016/ICT02/MMU/02/2 Malaysia and Shanghai 'The Belt and Road' Young Scholar Exchange Grant (17510740100).

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